

SUMMARY

Software Engineering student with hands-on, full-stack experience building production-style applications across web, mobile, and desktop platforms (React, Node.js, Flutter, Firebase, C++/Qt). Skilled at architecting multi-layer, multi-role systems and implementing secure authentication (JWT, RBAC). Strong problem-solving foundation from competitive programming, with a track record of mentoring 100+ students in algorithmic thinking.

EDUCATION

• Bachelor of Computer and System Engineering

Capital University

2023–2028

GPA: 3.8/4.00

PROJECTS

• UniSystem - Integrated University Management System

2026

Full-stack MERN platform unifying student, doctor, admin, and academic affairs services with embedded systems and AI.

- Architected and developed a comprehensive full-stack system using the MERN stack (MongoDB, Express, React, Node.js), unifying student registration, courses, lectures, and administration into a single platform.
- Implemented secure authentication with JWT tokens and role-based access control (RBAC) supporting Students, Doctors, Admins, and Academic Affairs staff.
- Designed an NFC-based smart attendance system enabling doctors to record attendance via NFC card scan, replacing manual roll-call and saving lecture time.
- Built a companion Flutter mobile app allowing students to check in via their phone's NFC reader as a backup to the physical card.
- Integrated an AI module that automatically generates lecture summaries and quizzes to help students study more effectively.
- Created an interactive discussion platform with real-time notifications for academic engagement.

• Gifting Platform - Multi-channel Gifting & Voucher Management System

2026

Full-stack comprehensive platform for sending gifts and vouchers across multiple channels with payment integration.

- Architected scalable 3-tier architecture: Flutter mobile app, React TypeScript web dashboard, and Node.js/Express backend with MongoDB.
- Integrated secure payment gateway (Stripe & local methods) with dynamic pricing and real-time transaction management.
- Implemented WhatsApp automation using Twilio API for gift notifications and recipient status updates.
- Built reusable gift templates system with category-based filtering, customization features, and template-category validation.
- Designed role-based authentication (RBAC) with JWT tokens supporting Customers, Admins, and Support teams.
- Leveraged Cloudinary for scalable media storage and Firebase for real-time notifications and data synchronization.

• Smart Dental Care System

2025

Full-stack AI-powered dental management ecosystem built with Flutter and Firebase.

- Developed a multi-role system (Patient, Doctor, Receptionist) featuring contactless QR check-in and automated digital invoicing via EmailJS.
- Integrated Google Generative AI (Gemini) to create a 24/7 smart dental assistant for instant patient support and medical advice.
- Built an interactive Tooth Chart for doctors to visually track dental conditions and integrated real-time chat for seamless clinic communication.
- Implemented "Family Mode" for centralized account management and utilized FL Chart for real-time clinic traffic and data visualization.

• Smart City Desktop Application

2025

Full-stack JavaFX desktop application unifying urban services, powered by Spring Boot and MongoDB.

- Built a role-based platform for Residents, Service Providers, and Admins with secure authentication and profile management.
- Developed a marketplace module for purchasing goods, with an initial focus on medicines, supporting role-aware ordering flows.
- Implemented an interactive map for locating nearby providers, enabling users to initiate live chat directly from the map.
- Engineered real-time messaging with Spring WebSocket and SockJS for instant, scalable communication between users.
- Integrated a weather dashboard, geolocation tools, and the Gemini API to power AI-assisted features.
- Designed the JavaFX 21 frontend against a Spring Boot REST API backend, with MongoDB for persistent data storage.

EXPERIENCE & TRAINING

• Trainee at CPC Community

CPC - Capital University

2024

- Developed strong foundational problem-solving skills and algorithmic thinking using C++.
- Practiced implementing core concepts including Time Complexity, Frequency Arrays, and Binary Search.
- Solved various problems focused on logic implementation through curated contests.

• Mentor at CPC Community

CPC - Capital University

2025

- Guided 100+ students in problem-solving techniques and algorithmic thinking using C++.

TECHNICAL SKILLS

Full-Stack Development: React, TypeScript, Node.js, Express, Flutter, Dart, Firebase, Spring Boot

Programming Languages: C++, Dart, JavaScript/TypeScript, Java, Python

Database & Cloud: MongoDB, Firebase (Firestore, Auth, Storage, Real-time DB), Cloudinary

Tools & Integrations: Git/GitHub, Linux, Stripe API, Twilio (WhatsApp), EmailJS, Vite, Qt, WebSocket

Core Concepts: RBAC, JWT Authentication, RESTful APIs, Real-time Communication, Payment Integration